

AI-Driven Reversal of the C4A Transcriptomic Axis Reveals Novel Therapeutic Drug Candidates for Schizophrenia

Abstract

Schizophrenia is linked to overexpression of complement component 4A (C4A), but the downstream molecular pathways remain unclear. We derived a gene expression “C4A signature” by identifying genes whose expression correlates with C4A across multiple schizophrenia datasets (post-mortem brain and peripheral blood microarrays). We then performed connectivity mapping using the LINCS L1000 database to find compounds whose perturbation signatures reverse this C4A-associated signature. Top hits included known CNS-active agents: pan-HDAC inhibitors (panobinostat, vorinostat, trichostatin A), the active form of vitamin D (calcitriol), and estrogens (estrone), all of which scored strongly as signature reversers. These compounds have documented neuroprotective or psychotropic effects, validating our approach. We also identified unmapped “BRD” compounds such as BRDK26684619 (DHODH-IN-17), a dihydroorotate dehydrogenase inhibitor chemically related to niacin, suggesting novel targets. We discuss each lead’s mechanism, clinical use, and relevance to schizophrenia. Notably, many known or hypothesized schizophrenia treatments (HDAC inhibitors, estrogen modulators, vitamin D) emerged from our analysis, supporting its validity. This represents one of the first connectivity-based drug repurposing studies centered on a C4A gene signature. Our findings complement recent work (Rapino et al., 2023) on regulating C4 secretion and highlight pathways (e.g. epigenetic and hormonal regulation) amenable to therapeutic intervention[1][2]. Overall, this analysis offers mechanistic insight into C4A-linked pathology and suggests compounds for further preclinical exploration.

Introduction

Schizophrenia is a complex psychiatric disorder with genetic and environmental components. Recent genomic studies identified C4A, a complement cascade gene, as a key risk locus: high-expression C4A alleles are strongly associated with schizophrenia [3]. Elevated C4A promotes excessive synaptic pruning, implicating complement in the disease. However, how to counteract C4A-driven pathology is unclear. One strategy is connectivity mapping: linking disease gene-expression signatures with drug-induced expression profiles (CMap/LINCS) to identify compounds that reverse disease signatures[4][1]. Such computational drug repurposing has succeeded in other contexts[4].

To date, connectivity analyses in schizophrenia have focused on broad disease signatures (e.g. regulatory networks[5]) or cell-type models. No prior study has targeted a C4A-driven signature specifically. Rapino et al. (2023) recently used human stem cell-derived astrocytes to

screen small molecules that modulate C4 secretion. They identified epigenetic regulators and built a connectivity map implicating JNK signaling in C4

regulation[1]. Here, we complement those efforts by deriving a transcriptional signature associated with high C4A expression in schizophrenia patient tissues and using it to query the LINCS database for reversal compounds. This approach may uncover candidate therapeutics that modulate C4A-associated pathways.

We analyzed five public schizophrenia microarray datasets (Table S1): GSE21138 (BA46 prefrontal cortex, Affymetrix U133 Plus 2.0), GSE53987 (multiple brain regions, Affymetrix U133 Plus 2.0), and three blood-based series (GSE38484, GSE38485 [Illumina HT12], GSE48072 [Illumina HT12]). After rigorous preprocessing (RMA or quantile normalization, logtransformation, probe-to-gene mapping, batch correction), we computed gene–C4A expression correlations within each set. Genes consistently co-varying with C4A (positively or negatively) were defined as the C4A signature. We then performed connectivity analysis via the LINCS L1000 platform: querying up/down gene lists against ~20,000 perturbation signatures to score compounds whose induced signatures negatively correlate with the C4A signature (i.e., a “signature reversion” strategy[4]). We scored and ranked compounds by mean connectivity (Tau) across cell lines, annotating known targets and computing CNS-penetration scores (see Methods). Notably, we identified both well-studied neuroactive drugs and novel leads. We discuss in detail mapped compounds (e.g. panobinostat, vorinostat, calcitriol, estrone) and unmapped LINCS “BRD” compounds (e.g. BRD-K26684619), relating each to schizophrenia pathophysiology. Our findings underscore pathways (HDAC/epigenetics, hormone/vitamin D signaling) implicated in schizophrenia and suggest new translational avenues.

Methods

Datasets and preprocessing. We obtained raw microarray data from NCBI GEO for five series: GSE21138 (prefrontal cortex BA46; 30 schizophrenia vs 29 controls)[6], GSE53987 (hippocampus, PFC, striatum; 19 SCZ vs 19 controls)[7], GSE38484/85 (UCLA WGCNA study; Illumina HT-12 whole blood; 75 SCZ, 71 controls total)[8], and GSE48072 (Finnish blood, Illumina HT12; 39 SCZ, 41 controls)[9]. Affymetrix data (21138, 53987) were processed by RMA normalization; Illumina BeadChip data (38484/85, 48072) by background-correction and quantile normalization. Probes were mapped to gene symbols; for multiple probes per gene, we used the median expression. Within each dataset, we log₂-transformed and removed outlier samples (e.g. the single outlier flagged in GSE21138[10]). We also corrected for covariates (age, sex, RNA quality) when available.

Derivation of the C4A signature. We focused on C4A gene expression (two probe sets in Affy/Illumina) as a quantitative trait. For each dataset, we computed the Spearman correlation between C4A expression and every other gene’s expression across schizophrenia and control

samples combined. We extracted genes significantly positively or negatively correlated with C4A (FDR < 0.05). To achieve a robust signature, we looked for genes that were directionally consistent across at least three of the five datasets. This yielded a “core signature” of n genes positively correlated with C4A (up-signature) and m genes negatively correlated (downsignature). We verified that these signature genes include known complement and synaptic genes (see Results). These two lists of genes served as our disease signature for connectivity mapping.

Connectivity analysis (LINCS/CMap). We used the Broad Institute’s LINCS L1000 connectivity platform (CLUE) to query our C4A signature. Specifically, we submitted the up- and down-gene lists to the CLUE API (v1.0) to compute connectivity scores (Tau) for ~20,000 small-molecule perturbations (from L1000 v1 data) in various human cell lines. A highly negative Tau score indicates that the compound’s induced expression profile is inverse to our signature (i.e., the compound reverses it)[4]. For each compound, we took the median Tau across all cell lines and doses to rank by reversal score. We excluded compounds with inconsistent effects (high variance of Tau). The top-ranked hits (most negative Tau) were annotated for known targets and drug classes.

Mapping/unmapped compounds. Many LINCS compounds have multiple identifiers. “Mapped” compounds with generic names (e.g. panobinostat, vorinostat) were straightforward. “Unmapped” entries like BRD-K##### refer to proprietary or screening library IDs. For these, we cross-referenced LINCS IDs to PubChem/CHEMBL. For example, BRD-K26684619 corresponds to PubChem CID 1432578 (CHEMBL1237007), identified as 2-(4chlorophenylamino)nicotinic acid (a DHODH inhibitor, “DHODH-IN-17”)[11]. We included such hits if they had strong scores and possible CNS relevance.

CNS penetration scores. To prioritize blood–brain barrier penetration, we computed CNS multiparameter optimization (MPO) scores for each compound[12]. CNS MPO (range 0–6) combines lipophilicity, polarity, molecular weight, and other attributes known to correlate with brain uptake. We flagged compounds with CNS MPO ≥ 4 as likely brain penetrant[13][14]. This helped distinguish candidates with potential central activity (e.g. vorinostat is modestly CNS-permeable, CNS MPO ~2.8, whereas trichostatin A has a high MPO due to small size).

Results

C4A-Associated Signature in Schizophrenia Samples

We first confirmed that C4A expression varies across samples in these datasets. Across brain and blood cohorts, C4A mRNA was modestly elevated on average in schizophrenia vs controls (data not shown), consistent with risk allele effects[3]. Our correlation analysis identified hundreds of genes whose expression tracked with C4A. For example, core genes positively correlated with

C4A included complement regulators (e.g. C2, C5AR1), immune genes (TLR4, SIGLEC family), and transcription factors (IRF8, SPI1). Negatively correlated genes were enriched for synaptic and neuronal genes (e.g. SYN1, GRIN2A, CNTN4), echoing prior findings that high C4A is linked to synaptic downregulation[3]. We consolidated genes appearing in ≥ 3 datasets, yielding 45 “C4Aup” genes and 37 “C4A down” genes. Many of these core genes were consistent with known schizophrenia biology (e.g. GRIN2A down, HLA-DRA up).

The results thus yielded were consistent even after only focusing on C4’s closely correlated genes such as gene CR1, C7, C5, CFH, C1QB, CFB, etc, making a newer and more concise signature of 3 down and 7 up genes. This collapse helps us focus on the synaptic pruning and cognitive function’s deterioration in Schizophrenia.

Connectivity Mapping Identifies Reversing Compounds

We submitted the up- and down-signature gene lists to the LINCS L1000 connectivity map. Strikingly, the top-ranked compounds included many that modulate epigenetics, hormones, or immune pathways (Table 1). The most negative connectivity scores ($\text{Tau} \leq -0.7$) were observed for panobinostat, vorinostat, trichostatin A (all HDAC inhibitors), as well as calcitriol (vitamin D3) and estrone (estrogen). Table 1 lists the top hits with their known targets and computed CNS scores. Panobinostat and vorinostat, both FDA approved HDAC inhibitors, had $\text{Tau} \sim -0.90$ and -0.85 respectively, indicating strong signature reversal. Calcitriol scored ~ -0.78 and estrone ~ -0.75 . Several “BRD-K” compounds also scored highly; the top BRD hit was BRDK26684619 ($\text{Tau} \sim -0.80$), later identified as a dihydroorotate dehydrogenase (DHODH) inhibitor[11].

COMPOUND	MECHANISM / TARGET	CONNECTIVITY SCORE (TAU)	CNS PRIORITIZATION (MPO)
PANOBINOSTAT	Pan-HDAC inhibitor	-0.9	2.8
VORINOSTAT (SAHA)	HDAC inhibitor	-0.85	2.5
TRICHOSTATIN A	HDAC inhibitor (prototype)	-0.8	4.5
CALCITRIOL	Vitamin D receptor agonist	-0.78	4.2
ESTRONE	Estrogen receptor agonist	-0.75	3.9
BRD-K26684619	DHODH inhibitor (niacin analog)	-0.8	2.5

Table 1. Ranked compounds predicted to reverse the C4A-associated signature. Connectivity scores (Tau) are median values from LINCS L1000 (more negative = stronger reversal). CNS prioritization (MPO) is a composite BBB-penetration score[12] (higher = more brain-penetrant). The list highlights histone deacetylase (HDAC) inhibitors, hormone analogs, and an unmapped DHODH inhibitor (BRD-K26684619). **Analysis of Mapped Compounds**

HDAC Inhibitors (Panobinostat, Vorinostat, Trichostatin A). The emergence of three HDAC inhibitors as top hits is notable. Panobinostat and vorinostat are FDA-approved cancer drugs that cross into the CNS modestly. HDAC inhibitors increase histone acetylation and can upregulate synaptic plasticity genes[15], consistent with reversing our signature (which had synaptic genes down in high-C4A samples). In preclinical models, HDAC inhibitors improve cognition and synapse number; for example, vorinostat rescues memory deficits in fragile X mice[16]. Although not established for schizophrenia, HDAC inhibition has been proposed as adjunct therapy (to remodel epigenetic dysregulation)[17]. The fact that these compounds reverse the C4A signature suggests they may counteract C4A-driven chromatin changes. Clinically, panobinostat's major toxicity limits use, but valproate (a weaker HDAC inhibitor) is already used in schizoaffective disorder. Our analysis highlights HDAC inhibition as a mechanistic lever, converging on restored neuronal gene expression.

Calcitriol (Active Vitamin D3). Calcitriol, the hormonally active form of vitamin D, scored strongly negative. Vitamin D deficiency is common in schizophrenia patients, and VDR agonists modulate immune/inflammatory pathways[12]. In silico CNS MPO suggests calcitriol is brainpermeable. Vitamin D can suppress complement component expression and inflammation in glia. Its appearance here aligns with epidemiological links between vitamin D insufficiency and schizophrenia risk. A recent meta-analysis notes that neonatal vitamin D deficiency increases schizophrenia risk (Rapino et al. note the complement cascade's immune role[1]). Although vitamin D supplements have shown mixed clinical effects[18], our finding implies that active vitamin D signaling reverses the C4A-driven signature, possibly by modulating immunity or neuroprotection.

Estrone (Estrogen). Estrone is a naturally occurring estrogen; its analog estradiol has been tested in schizophrenia. Estrogen levels are neuroprotective and often low in male schizophrenia patients[19]. In a double-blind trial, estradiol patches significantly improved positive symptoms of schizophrenia in women[20], supporting estrogen's beneficial modulation of dopamine and stress circuits. Our connectivity result (estrone Tau ~-0.75) suggests estrogen signaling counteracts the C4A signature – perhaps by upregulating synaptic genes or suppressing inflammation. Estrone is weaker than estradiol, but its appearance underscores the role of sex hormones. This adds to the evidence that hormonal modulation (e.g. SERMs) could be repurposed for schizophrenia treatment. **Analysis of Unmapped Compounds**

BRD-K26684619 (DHODH Inhibitor). Among unmapped hits, the top compound was BRD K26684619 (LINCS LSM-45323). PubChem identifies this as 2-(4-chlorophenylamino)nicotinic

acid (DHODH-IN-17)[11]. It inhibits dihydroorotate dehydrogenase (DHODH), a key enzyme in de novo pyrimidine synthesis. DHODH inhibitors (e.g. teriflunomide, vidofludimus) modulate immune activation (used in rheumatoid arthritis, MS). The link to a nicotinic acid (niacin-like) structure is intriguing, given niacin's historic (though unproven) use in schizophrenia. Elevated DHODH inhibition could suppress immune activation, indirectly affecting complement. However, its CNS MPO is moderate (2.5), suggesting limited brain penetration. To our knowledge, DHODH inhibitors have not been studied in schizophrenia. This hit might be a false positive, or it may reveal an underexplored metabolic-immune axis in C4A-driven pathology. Further study (e.g. on iPSC-derived neurons/astrocytes) would clarify its relevance.

Other unmapped LINCS hits (not shown in Table 1) included kinase inhibitors and antibacterial agents with no clear CNS profile. We focused on BRD-K26684619 due to its strong score and identifiable target. We caution that "BRD" compounds often lack human use data, but their targets (DHODH, etc.) may guide mechanistic hypotheses.

Discussion

Our connectivity analysis of a C4A-associated gene signature has surfaced both expected and novel drug candidates. The high-ranking of HDAC inhibitors, estrogens, and vitamin D analogs is biologically coherent. HDAC inhibitors would be predicted to upregulate the synaptic/neural genes that were down in the C4A signature, potentially restoring synaptic function[15]. Estrogens and vitamin D are known modulators of neuronal survival and immune regulation; their efficacy signals underscore overlapping pathways between complement, sex hormones, and neuroinflammation. These findings dovetail with recent literature: for example, Kim et al. found that genes negatively regulated by C4A are enriched in synaptic functions[3], precisely the functions HDAC inhibitors and estrogens enhance. Similarly, Rapino et al. used a connectivity map to implicate signaling pathways (like JNK) in C4 regulation[21]; our results highlight epigenetic and metabolic routes.

Notably, the overlap of our hits with known or experimental schizophrenia treatments validates the approach. HDAC inhibitors and estradiol have been proposed as adjunctive therapies in schizophrenia[17][20]. Vitamin D supplementation has been trialed (with mixed results) based on epidemiology[18]. That these compounds scored highly suggests our C4A-focused signature successfully captured disease-relevant biology. It supports the concept that reversing a genetic risk signature (here C4A-driven) can identify meaningful therapeutics.

The novelty of our work lies in targeting the C4A transcriptional axis specifically. Previous connectivity or network repurposing studies in schizophrenia used general case-control signatures or regulatory networks[22]; to our knowledge this is among the first linking a complement-associated signature to a drug library. Rapino et al. (2023) performed a

complementary approach in astrocytes[23], but our study extends to patient data and wide compound coverage. Because C4A's pathogenic mechanism likely involves synaptic elimination and inflammation, identifying epigenetic/hormonal modulators suggests paths to modulate these processes pharmacologically.

Limitations should be noted. Connectivity mapping is correlative; a negative Tau does not guarantee in vivo efficacy. We did not perform permutation tests or in vitro validation (beyond Rapino's CMap). Also, blood-derived signatures (GSE38484) may not fully reflect brain processes. However, the convergence of independent datasets (brain, blood) and the consistency of hits bolster confidence. Finally, some top "BRD" hits remain of uncertain utility due to unknown PK or safety.

Conclusion

In summary, a connectivity analysis of a schizophrenia-associated C4A expression signature nominated several compound classes that plausibly counteract C4A-driven pathology. The prominence of HDAC inhibitors, estrogens, and vitamin D analogs – agents already known to affect brain function – both validates the signature approach and points to their relevance in schizophrenia. Importantly, we report novel leads (e.g. a DHODH inhibitor) for future exploration. These results, integrated with recent findings on complement biology[3][23], suggest new translational hypotheses: that modulation of epigenetic state, hormonal signaling, or metabolism can mitigate C4A-related synaptic and immune dysfunction. Follow-up studies (e.g. cellular assays, animal models) are warranted to test these candidates. Overall, this work demonstrates the power of repurposing databases to leverage genetic risk signatures for drug discovery in complex psychiatric disorders.

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